

Syllabus

Modern Simulation Software Development · Summer 2026

1 Course Information

Lecturers	Dr. Lambert Theisen , Dr. Georgii Oblapenko
ECTS	5
Format	2h lecture + 1h tutorial/week (tutorials merged to 2h biweekly)
Examination	Oral exam or project-based evaluation (TBD)
Module	Current Topics in Computational Science and Engineering
Registration	RWTHOnline (course 11.00153) – opens 11 March 2026

2 Background Literature

- D. A. Kopriva (2009). *Implementing Spectral Methods for Partial Differential Equations: Algorithms for Scientists and Engineers*. Springer.
- A. Logg, K. A. Mardal & G. Wells, Eds. (2012). *Automated Solution of Differential Equations by the Finite Element Method: The FEniCS Book*. Springer.
- R. C. Smith (2024). *Uncertainty Quantification: Theory, Implementation, and Applications*. SIAM.
- I. D. Boyd & T. E. Schwartzentruber (2017). *Nonequilibrium Gas Dynamics and Molecular Simulation*. Cambridge University Press.

3 Links

- [ACoM Website](#)
- [Course Announcement](#)
- [Moodle Page](#)